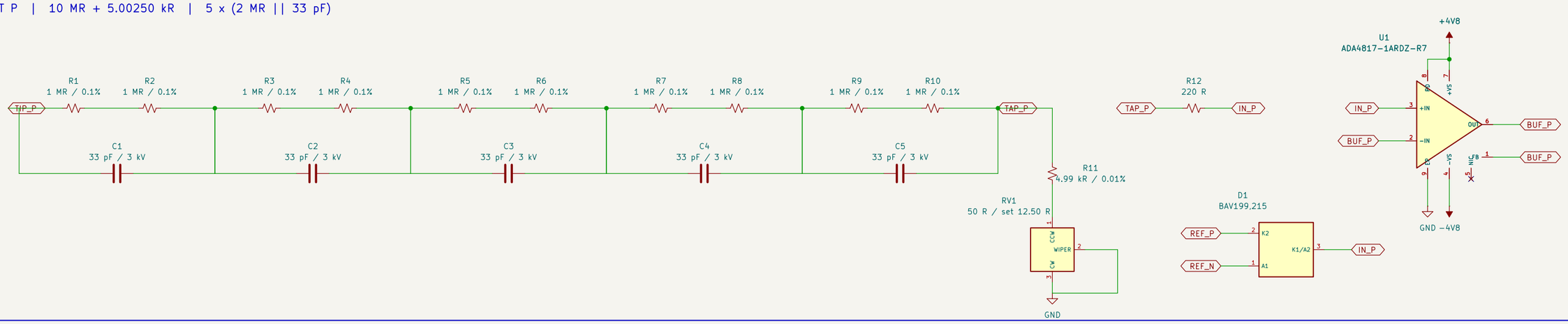
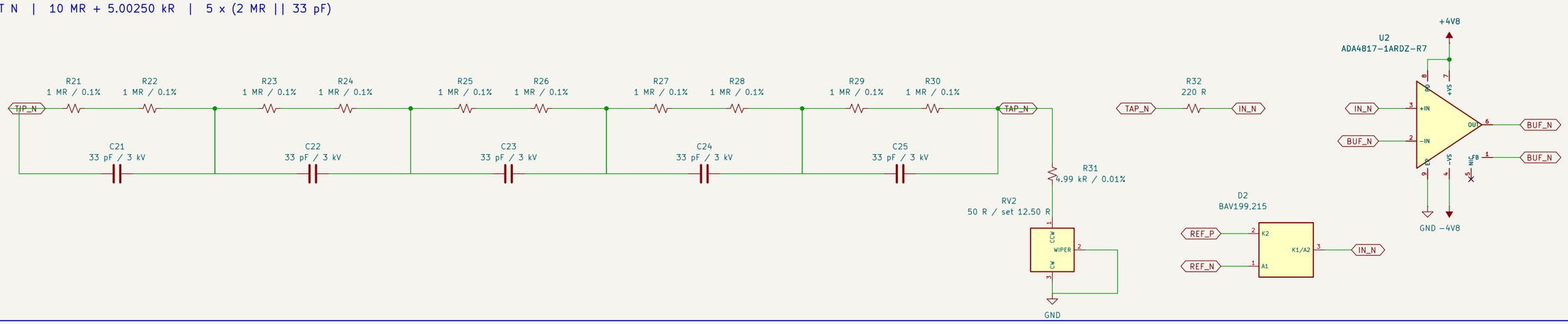
2000:1 INPUTS / matched geometry, fixed grounded shield

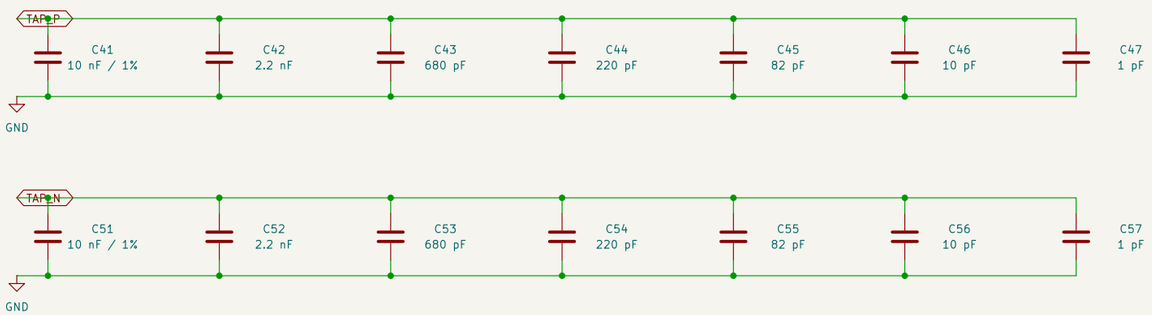
INPUT P | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)



INPUT N | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)



AC COMPENSATION / C0G selection pads; trim with shield fitted



Starting Csum = 13.193 nF per leg. Select on the bench.
Do not assume nominal values deliver 60 dB CMRR.
Local decoupling for U1/U2 is on the power sheet.
TIP_P / TIP_N: captive HV leads, no SMA inputs.

signal

File: signal.kicad_sch

power

File: power.kicad_sch

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /

File: differential−probe.kicad_sch

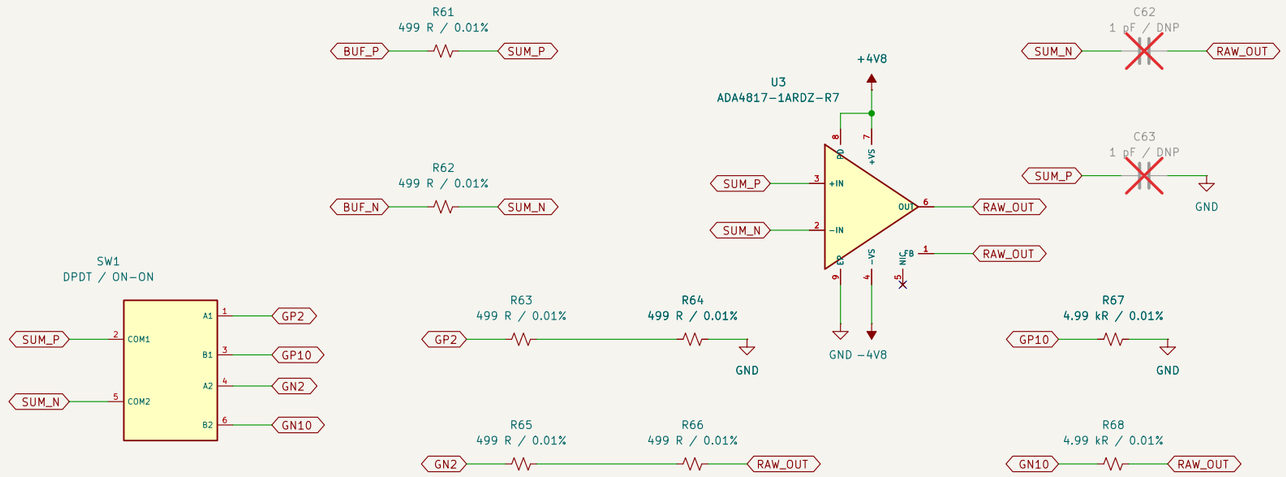
Title: Differential probe / compensated inputs

Size: A3	Date: 2026−10−20	**Rev: A prototype**
KiCad E.D.A. 10.0.4		Id: 1/3

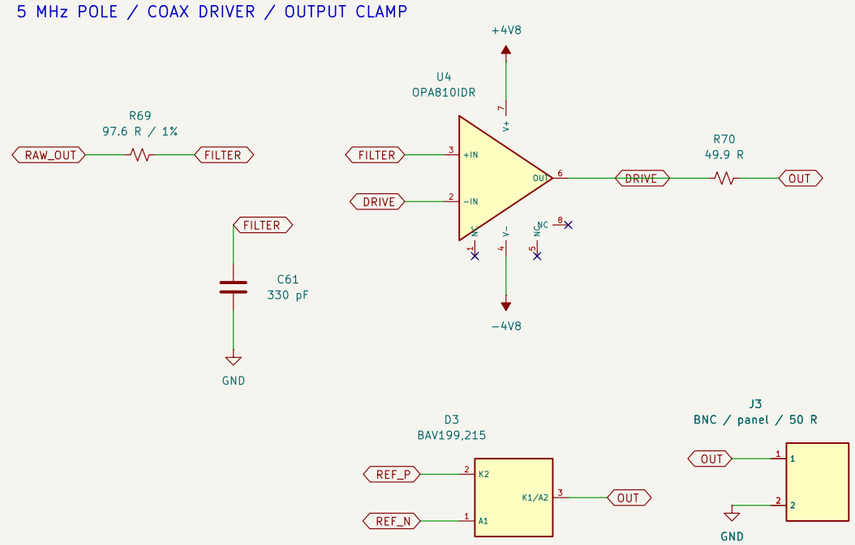
1000:1 / 200:1 OUTPUT / scope input = 1 MR

PRECISION DIFFERENCE AMPLIFIER / G = 2 or 10

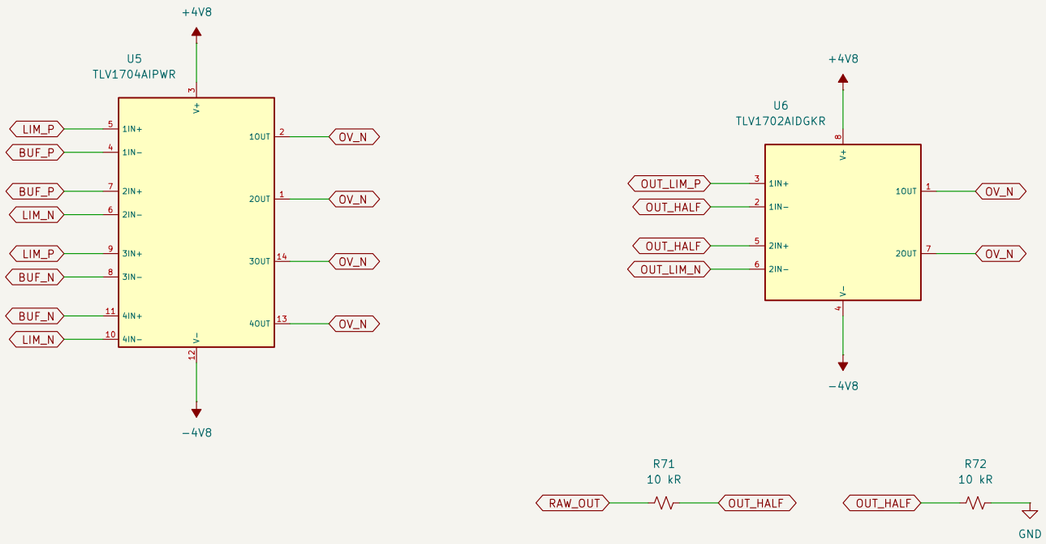
SW1: 2–1 / 5–4 = x2; 2–3 / 5–6 = x10.
Change range with inputs de-energized. Keep switch at U3.



5 MHz POLE / COAX DRIVER / OUTPUT CLAMP

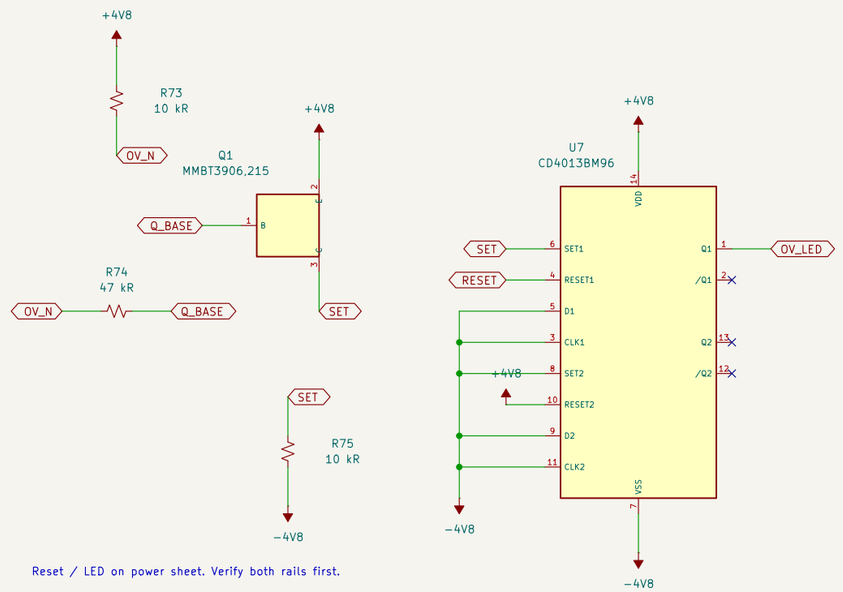


OVERLOAD DETECTION / either input exceeds +/−1.60 V or raw output +/−3.0 V



TLV170x: 560 ns typical delay. Indicator is not a fast peak detector.
OUT clamp is downstream of U3: amplifier recovery still needs qualification.

LATCHED OVERLOAD / RESET



Reset / LED on power sheet. Verify both rails first.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /signal/
File: signal.kicad_sch

Title: Differential probe / gain, output and overload

Size: A3	Date: 2026-10-20	**Rev: A prototype**
KiCad E.D.A. 10.0.4		Id: 2/3

BATTERY POWER / scope shield defines circuit ground

TWO 9 V BATTERIES / +/−4.8 V LOW-NOISE RAILS

BT1
9 V battery clip
BAT_P
GND

BT2
9 V battery clip
GND
BAT_N

SW2
POWER / DPDT / 5 A
BAT_P
BAT_N
GND
SW_P
SW_N

+BAT
D4
SS14
SW_P

D5
SS14
SW_N
−BAT

U8
ADP7142AUJZ-R7
+BAT
+BAT
VIN
EN
VOUT
ADJ
GND
+4V8
ADJ_0

U9
ADP7182AUJZ-R7
−BAT
−BAT
VIN
EN
VOUT
ADJ
GND
−4V8
ADJ_1

C71
10 uF / 25 V
+BAT
GND

C73
10 uF / 25 V
−BAT
GND

R81
30 kR
+4V8
ADJ_0

R83
29.4 kR
−4V8
ADJ_1

C72
10 uF / 25 V
+4V8
GND

R82
10 kR
ADJ_0
GND

C74
10 uF / 25 V
−4V8
GND

R84
10 kR
ADJ_1
GND

SHUNT CLAMPS / COMPARATOR THRESHOLDS

+4V8
R85
1 kR
REF_P

U10
TL431BIDBZR
REF_P
REF_P
K
REF
A
GND

R87
5.62 kR
REF_P
LIM_P

R88
10 kR
LIM_P
GND

R89
6.65 kR
REF_P
OUT_LIM_P

R90
10 kR
OUT_LIM_P
GND

−4V8
R86
1 kR
REF_N

U11
TL431BIDBZR
REF_N
GND
K
REF
A
REF_N

R91
5.62 kR
REF_N
LIM_N

R92
10 kR
LIM_N
GND

R93
6.65 kR
REF_N
OUT_LIM_N

R94
10 kR
OUT_LIM_N
GND

LOCAL DECOUPLING / RAIL REVERSAL DIODES

+4V8
U1 U2 U3 U4 U5 U6 U7
C81 C83 C85 C87 C89 C91 C93
100 nF / 50 V
GND
C82 C84 C86 C88 C90 C92 C94
100 nF / 50 V
−4V8

D6
SS14
+4V8
GND

D7
SS14
GND
−4V8

RESET / STATUS / POWER VALIDATION

SW3
RESET / NO
+4V8
RESET

R95
100 kR
RESET
−4V8

+4V8
C95
100 nF / 50 V
RESET

D8
1N4148W,115
−4V8
RESET

R96
10 kR
OV_LED
LED_A

D9
RED / OVERLOAD
LED_A
−4V8

Two 9 V alkaline batteries; center tap = GND.
Verify rails, reset, then connect DUT.
No charging / USB connection during measurement.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /power/
File: power.kicad_sch

Title: Differential probe / battery supply and references

Size: A3	Date: 2026-10-20	**Rev: A prototype**
KiCad E.D.A. 10.0.4		Id: 3/3